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AKASH INSTITUTE OF ENGINEERING AND TECHNOLOGY

Affiliated to Visvesvaraya Technological University, Belagavi and Approved by AICTE, New Delhi
Devanahalli, Bengaluru- 562110

DEPARTMENT OF CSE
IMPORTANT QUESTIONS

Course Name	Operating Systems	**Course Code**	1BCS304
Year/Semester/Section	2nd /3rd /A,B,C	**Academic Year**	2026-27

Q.No.	Questions	Marks	CO
1	Define an Operating System. Explain dual-mode operation (user mode and kernel mode) with a neat diagram	5	CO 1
2	Differentiate between: • Multiprogramming and multitasking • Multiprocessor systems and clustered systems.	5	CO 1
3	Explain the services provided by an operating system for users and for efficient system operation.	5	CO 1
4	Explain the layered approach to operating system structure with a supporting diagram	5	CO 1
5	Explain the different components and functions of an operating system, including process management, memory management, storage management, protection and security.	5	CO 1
6	Explain operating-system operations and system programs. Discuss operating-system design and implementation, system generation, and system boot.	5	CO 1
7	Define a process. Explain the different process states with a neat state-transition diagram and describe the Process Control Block (PCB).	5	CO 2
8	What is Inter-Process Communication (IPC)? Explain shared-memory and message-passing models. Differentiate between direct and indirect communication	5	CO 2
9	Define a thread. Explain the different multithreading models — many-to-one, one-to-one and many-to-many — with neat diagrams. List the benefits of multithreaded programming.	5	CO 2
10	Explain CPU scheduling criteria: CPU utilization, throughput, turnaround time, waiting time and response time	5	CO 2
11	What is context switching? Explain the steps involved in context switching and its effect on system performance	5	CO 2
12	Explain the different threading issues in multithreaded programming, including fork/exec semantics, signal handling, thread cancellation and thread pools	5	CO 2
13	Explain the concept of process synchronization. What is the critical section	5	CO

	problem? State its requirements.		2
14	Explain Peterson's solution to the critical section problem with an algorithm	5	CO 3
15	What are semaphores? Explain binary and counting semaphores with examples	5	CO 3

Sl. No	Description
CO1	Recall the structure, functions, and services of an operating system.
CO2	Explain process management concepts, CPU scheduling algorithms, and process synchronisation mechanisms.
CO3	Apply appropriate CPU scheduling algorithms and process synchronisation techniques to solve operating system problems.
CO4	Apply memory management techniques and deadlock-handling strategies to ensure efficient resource utilization.
CO5	Explain file system organisation and secondary storage management techniques.